

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS/POST Error Message	Hardware or Configuration Issue
CMOS Checksum Error – Defaults Loaded	CMOS battery is weak or BIOS settings have been reset.
Keyboard Error or No Keyboard Present	Keyboard is disconnected, faulty, or USB/PS2 port issue.
Disk Boot Failure, Insert System Disk	No bootable operating system or hard drive/SSD not detected.
CPU Fan Error	CPU cooling fan is not connected, faulty, or spinning too slowly.
Memory Test Failed	RAM is faulty, improperly installed, or incompatible.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Meaning:

For a typical **AMI BIOS**, **3 short beeps** usually indicate a **memory (RAM) failure** or a problem with the first 64 KB of system memory.

Troubleshooting Steps:

1. Remove the RAM modules, clean the contacts if necessary, and firmly reseal them in the correct memory slot.
2. Test each RAM stick individually or replace it with a known working RAM module to identify a faulty memory stick or slot.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

BIOS Version/Date	LENOVO R0SET49W (1.33), 12/27/2019
SMBIOS Version	3.0
Embedded Controller Version	1.33
BIOS Mode	UEFI
BaseBoard Manufacturer	LENOVO
BaseBoard Product	20LJS13E01
BaseBoard Version	SDK0J40697 WIN

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

Meaning

This message indicates that the BIOS detected the saved CMOS settings are invalid or have changed unexpectedly. The BIOS automatically loads the default settings because the stored configuration could not be verified. This commonly happens due to a weak or dead CMOS battery or after the CMOS has been cleared.

How to Resolve

1. Enter the BIOS Setup.
2. Check and correct the **date, time, and BIOS settings**.
3. Save the settings and restart the computer.
4. If the error appears again, replace the **CMOS battery (CR2032)** on the motherboard.
5. After replacing the battery, reconfigure the BIOS settings and save them.
6. If the problem continues, update the BIOS (if required) or inspect the motherboard for hardware faults.

